

DROXY-50 Water Soluble Powder

Droxy-50 appears to be a yellowish crystalline powder packed in aluminium bag with 120 micron thickness.

Composition:

Each kg contains

Doxycycline Hyclate

500 g

Pharmacodynamics:

Tetracyclines are broad-spectrum bacteriostatic agents that inhibit protein synthesis by binding reversibly to receptors of the 30 S ribosomal subunit of susceptible microorganisms. The binding of a tetracycline to the subunit blocks the binding of the aminoacyl tRNA to the acceptor site on the mRNA-ribosomal complex and prevents the addition of new amino acids to the peptide chain, inhibiting protein synthesis. Tetracyclines must enter the target cell to be effective. Uptake appears to depend on passive diffusion and active transport, with the exception of doxycycline, which enters the cell by passive diffusion. Susceptible cells concentrate the antibiotic; resistant strains appear to carry an R-factor that inhibits uptake of drug.

Pharmacokinetics:

After oral administration to pigs, doxycycline is substantially absorbed from the gastrointestinal tract. The binding rate to plasma proteins is 93%. It is widely distributed in the organisms; at the steady state, the volume of distribution (VSS) is 1.2 L/kg. Doxycycline is not metabolised to any significant extent and it is excreted primarily in faeces, mostly in a microbiologically inactive form. The elimination half-life was reported to be 4-4.2 hours.

Pharmacokinetics of doxycycline after single oral administration to chickens and turkey is characterised by a quite rapid and substantial absorption from the gastrointestinal tract providing peak plasma concentrations between 0.4 and 3.3 hours (chickens), 1.5 and 7.5 hours (turkeys) depending on age and the presence of food. The drug is widely distributed in the organism with Vd values close to or greater than 1, and exhibits an elimination half-life of 4.8 to 9.4 hours (chickens), 7.9 to 10.8 hours (turkeys). The protein binding ratio at therapeutic plasma concentrations is in the range of 70-85%. The bioavailability in chickens may vary between 41 and 73% while in turkeys between 25 and 64%, also depending on the age and feeding. The presence of food in the gastrointestinal tract determines a lower bioavailability compared to that obtained in the fasted state.

Indications:

Pigs: Treatment of clinical respiratory infections caused by *Mycoplasma hyopneumoniae* and *Pasteurella multocida* susceptible to doxycycline.
Chickens and turkeys: Treatment of clinical respiratory infections associated with *Mycoplasma gallisepticum* susceptible to doxycycline.

Directions for Use:

In pigs and chickens, 20.0 mg doxycycline per kg of body weight daily (equivalent to 44 mg product per kg of body weight), administered in the drinking water for 5 consecutive days.

In turkeys, 25 mg doxycycline per kg of body weight daily (equivalent to 55 mg product per kg of body weight), administered in the drinking water for 5 consecutive days.

The product should be administered continuously in the drinking water during the whole period of treatment. The medicated water should be the only source of drinking water, throughout the treatment period. To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.

Mode of Administration: Oral Water soluble powder

Contraindication:

Do not use in case of hypersensitivity to the active substance or to any of the excipients.

Do not use when tetracycline resistance has been detected in the herd/flock due to the potential for cross resistance.

Do not use in animals with hepatic dysfunction.

Warning and Precautions:

The use of doxycycline in animals with an impaired liver function should be avoided.

Do not combine with bactericidal antimicrobials such as penicillins or cephalosporins.

Solubility of doxycycline is pH dependent. Precipitation will occur in an alkaline solution. This product should therefore not be mixed with other products.

Special warnings for each target species

The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of drinking water, animals should be treated parenterally.

Do not administer to poultry whose eggs are intended for human consumption.

Special precautions for use in animals

Inappropriate use of the product may increase the prevalence of bacteria resistant to tetracycline due to the potential for cross resistance.

Use of the product should be based on susceptibility testing and take into account official and local antimicrobial policies.

The safety of the product has not been established in piglets before weaning.

Interactions with Other Medicaments:

Do not administer concurrently with feed overloaded with polyvalent cations such as Ca^{2+} , Mg^{2+} , Zn^{2+} and Fe^{3+} because the formation of doxycycline complexes with these cations is possible. Do not administer together with antacids, kaolin and iron preparations as tetracyclines are bacteriostatic antimicrobials, do not administer in conjunction with bactericidal antibiotics like beta-lactams. It is advised that the interval between administrations of other products containing polyvalent cations should be 1-2 hours because they limit the absorption of tetracycline.

Doxycycline increases the action of anticoagulants.

Statement on Usage During Pregnancy and Lactation:

Laboratory studies in rats and rabbits have not produced any evidence of a teratogenic, foetotoxic, maternotoxic effects. The safety of the product has not been established in pregnant or lactating sows. The use is not recommended during pregnancy and lactation. Do not use in birds in lay and within 4 weeks before the onset of the laying period.

Adverse Effects / Undesirable Effects:

As for all tetracyclines, on rare occasions allergic reactions and photosensitivity may occur. If suspected adverse reactions occur, treatment should be discontinued. If you notice any serious effects or other effects, please inform your veterinary surgeon.

Overdose and Treatment::

During the target animal tolerance study, no adverse effect was observed even at the fivefold therapeutic dose administered for two times the recommended duration in either target animal species. If suspected toxic reactions do occur due to extreme overdose, the medication should be discontinued and appropriate symptomatic treatment should be initiated if necessary.

Withdrawal Period (s):

Meat and offal of pigs: 4 days.

Meat and offal of chickens: 5 days.

Meat and offal of turkeys: 12 days.

Not permitted for use in laying birds producing eggs for human consumption.

Storage Condition:

Store below 30 ° C. Protect from heat and sunlight exposure.

Keep medicine out of reach of children (Jauhkan ubat dari kanak-kanak).

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

Shelf life packaged for sale: 3 years

Shelf-life after dilution: 1 day

Packaging Available: 100g, 200 g & 500 g

Date of revision: 09-11-2020

Product Holder/ Manufacturer:

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